VIDYAPEETH

BATCH CODE: 12-AN-181-MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNIT & MEASUREMENT

Lecture No. 14

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Today's Goal

Subtopic .. Error Analysis

Subtopic ..

Subtopic ..

Q. $r = (2.1 \pm 0.5)\,cm$

Calculate Volume of the sphere with Error Limits

Sol

$$\boxed{V = \frac{4}{3}\pi r^3}$$

$$V = \frac{4}{3} \times \frac{22}{7} \times 2.1 \times 2.1 \times 2.1$$

$$= 38.8\ cm^3$$

$$\frac{\Delta V}{V} = \frac{3\Delta r}{r} = 3 \times \frac{0.5}{2.1} = 0.71$$

$$\frac{\Delta V}{V} = 0.71 \Rightarrow \Delta V = 0.71 \times 38.8$$

$$= 27.7$$

$$\boxed{V = (38.8 \pm 27.7)\ cm^3}$$

Q. $Z = \dfrac{A^4 B^{1/3}}{C D^{3/2}}$

Find Relative Error in Z if % error in A is 4%, B is 2%, C is 3%, D is 1%

$$100 \times \frac{\Delta Z}{Z} = \left[4\frac{\Delta A}{A} + \frac{1}{3}\frac{\Delta B}{B} + 1\frac{\Delta C}{C} + \frac{3}{2}\frac{\Delta D}{D}\right] \times 100$$

$$\frac{\Delta Z}{Z} \times 100 = (4 \times 4) + \left(\frac{1}{3} \times 2\right) + 3 + \left(\frac{3}{2} \times 1\right)$$

$$= 16 + \frac{2}{3} + 3 + \frac{3}{2} = \frac{96 + 4 + 18 + 9}{6}$$

$$\frac{\Delta Z}{Z} \times 100 = \frac{127}{6} = 21.11$$

$$\boxed{\frac{\Delta Z}{Z} = \frac{127}{600} = 0.211}$$

% error in A = 4%

$$\frac{\Delta A}{A} \times 100 = 4$$

$$\frac{\Delta A}{A} = \frac{4}{100}$$

$$100 \times \frac{\Delta Z}{Z} = \left[4\left(\frac{\Delta A}{A}\right) + \frac{1}{3}\frac{\Delta B}{B} + \frac{\Delta C}{C} + \frac{3}{2}\frac{\Delta D}{D}\right] \times 100$$

$$\frac{\Delta Z}{Z} = \frac{(4 \times 4) + \frac{1}{3}(2) + 3 + \frac{3}{2}(1)}{100}$$

$$= \frac{127}{600} = 0.21$$

Q. $r = (2.1 \pm 0.5)\,cm$

Calculate Volume of the sphere with Error Limits

Sol

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3} \times \frac{22}{7} \times 2.1 \times 2.1 \times 2.1$$

$$= 38.8\ cm^3$$

$$\frac{\Delta V}{V} = \frac{3\Delta r}{r} = 3 \times \frac{0.5}{2.1} = 0.71$$

$$\frac{\Delta V}{V} = 0.71 \Rightarrow \Delta V = 0.71 \times 38.8$$

$$= 27.7$$

$$V = (38.8 \pm 27.7)\ cm^3$$

Q. $Z = \frac{A^4 B^{1/3}}{C D^{3/2}}$

Find Relative Error in Z if % error in A is 4%, B is 2%, C is 3%, D is 1%

Q. $P = \frac{A^2 \sqrt{B}}{C}$

% error in A = 1%, B = 2%, C = 4%

Find Rel. Error in P.

$$\frac{7}{100}$$

Q.

$u = (10 \pm 0.1)\ cm$

$v = (10 \pm 0.1)\ cm$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}} = \frac{u+v}{uv}$$

$$\boxed{f = \frac{uv}{u+v}}$$

$$\boxed{f = \frac{10 \times 10}{10+10} = 5\ cm}$$

$\Delta f = ?$

$$\boxed{f = \frac{uv}{(u+v)}} = \frac{uv}{w}$$

$$\frac{\Delta f}{f} = \frac{\Delta u}{u} + \frac{\Delta v}{v} + \left[\frac{(\Delta u + \Delta v)}{u+v}\right]$$

$$= \frac{0.1}{10} + \frac{0.1}{10} + \frac{0.1+0.1}{20}$$

$$\frac{\Delta f}{f} = \frac{1}{100} + \frac{1}{100} + \frac{1}{100}$$

$$\Delta f = \frac{3}{\cancel{100}_{20}} \times \cancel{5} \Rightarrow \Delta f = \frac{3}{20}\ cm = 0.15$$

$$\boxed{f = (5 \pm 0.15)\ cm}$$

ये गलत है।
ये नहीं करना है

i) Here in the term $\left(\frac{\Delta u + \Delta v}{u+v}\right)$; errors Δu & Δv are dependent on each other ; which is not correct. Absolute errors should must be independent.

ii) Eg :- Equivalent Resistance in Parallel connection

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \boxed{R_{eq} = \frac{R_1 R_2}{R_1 + R_2}}$$

$$\frac{\Delta R_{eq}}{R_{eq}} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} + \left(\frac{\Delta R_1 + \Delta R_2}{R_1 + R_2}\right)$$

Not Correct

iii) Instead we'll use Differentiation Method :-

$$\boxed{\frac{\Delta f}{f^2} = \frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}}$$

$$\boxed{\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}}$$

Again :-

$f = 5\,cm$

$u = 10 \pm 0.1$

$v = 10 \pm 0.1$

NCERT

$$\boxed{\frac{\Delta f}{f^2} = \frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}}$$

$$\frac{\Delta f}{25} = \frac{0.1}{100} + \frac{0.1}{100}$$

$$\Delta f = \cancel{25}\left[\frac{1}{\cancel{500}\,20}\right]$$

$$= \frac{1}{20}\,cm = 0.05\,cm$$

$f = (5 \pm 0.05)\,cm \rightarrow \underline{4.95 \text{ to } 5.05}$

$f \pm \Delta f$

★★★★★

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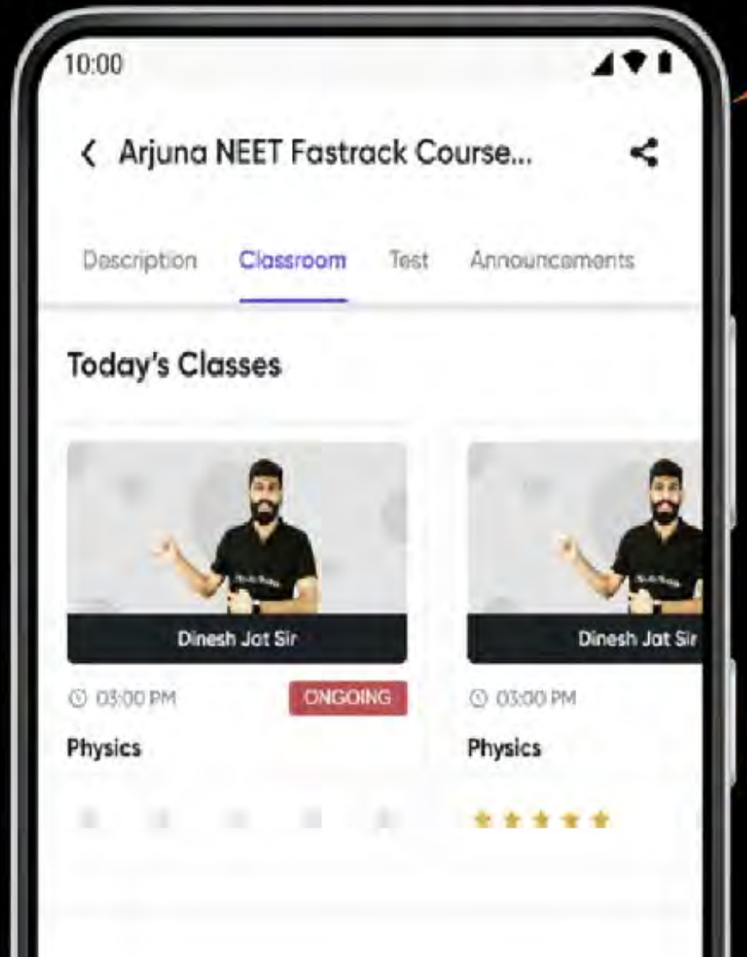
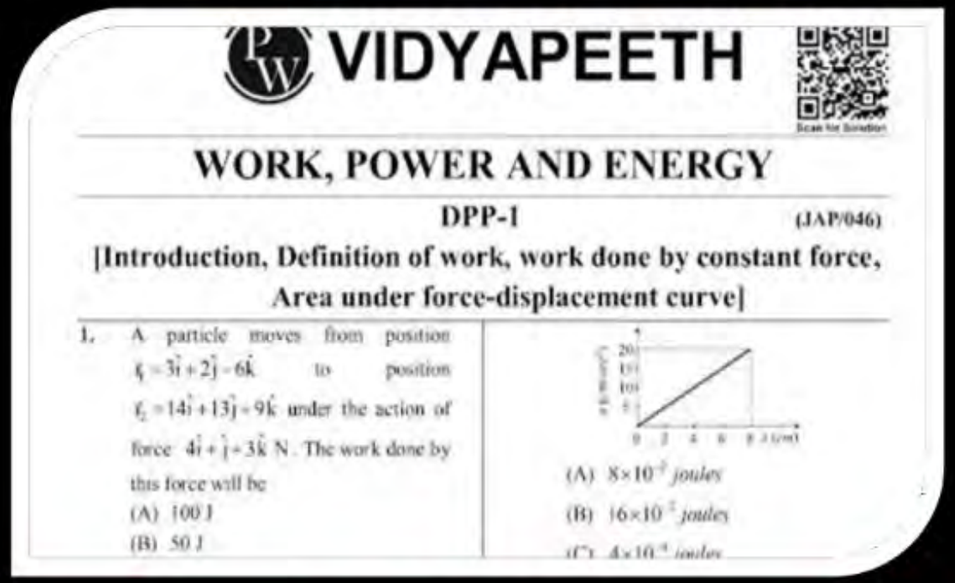


Solve the DPP and check Solution



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